

LESSON 6-3 • SESSION 1 • GUIDED NOTES**Writing Repeated Factors as Powers**

Learning target – I can write repeated multiplication as a power and name its base and exponent. · 6.EE.A.1

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
power <i>potencia</i>	A short way to write the same _____ multiplied again and again.	9^4
base <i>base</i>	The number that is _____ as a factor.	In 9^4 , the base is 9
exponent <i>exponente</i>	The raised number that tells how many _____ the base is a factor.	In 9^4 , the exponent is 4

KEY IDEA*complete each sentence*

- The _____ is the repeated factor.
- The _____ tells how many times the base is used as a factor.
- The exponent counts factors, not _____ signs.

WORKED EXAMPLE*fill in the blanks as we solve*Write $2 \times 2 \times 2 \times 2$ as a power. Name the base and the exponent.**Step 1** The repeated factor is _____. That is the base.**Step 2** Count the factors: there are _____. That is the exponent.**Step 3** Write the base with a raised exponent: 2^4 .**Answer** 2^4 , read “2 to the _____ power”**YOU TRY***use the example as a model***1.** Write each product as a power.

a. $6 \times 6 \times 6 =$ _____

b. $1/4 \times 1/4 =$ _____

2. Write 10^4 in expanded form. How many factors does it have?*Show your work.***Factors** _____**SUM IT UP***in your own words*In 7^4 , the 4 tells me _____ because _____.

LESSON 6-3 • SESSION 2 • GUIDED NOTES

Finding the Value of a Power

Learning target – I can find the value of a power by multiplying the base the number of times the exponent shows. ·
6.EE.A.1

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
evaluate <i>evaluar</i>	Find the one _____ an expression equals.	$2^3 = 8$
squared <i>al cuadrado</i>	To the 2nd power; use the base as a factor ____ times.	$4^2 = 4 \times 4 = 16$
cubed <i>al cubo</i>	To the 3rd power; use the base as a factor ____ times.	$4^3 = 4 \times 4 \times 4 = 64$

KEY IDEA*complete each sentence*

- To evaluate a power, write the _____ form and multiply one factor at a time.
- $4^3 = 4 \times 4 \times 4 = 64$, not $4 \times 3 = \underline{\hspace{1cm}}$.
- Fraction and decimal bases work the _____ way: $(1/2)^2 = 1/4$.

WORKED EXAMPLE*fill in the blanks as we solve*Find the value of 6^3 .**Step 1** Expand: $6^3 = 6 \times 6 \times \underline{\hspace{1cm}}$.**Step 2** Multiply the first two: $6 \times 6 = \underline{\hspace{1cm}}$.**Step 3** Multiply by the last 6: $36 \times 6 = \underline{\hspace{1cm}}$.**Answer** $6^3 = \underline{\hspace{1cm}}$, not $6 \times 3 = 18$ **YOU TRY***use the example as a model***1.** Find the value of each power.

a. $9^2 = \underline{\hspace{2cm}}$

b. $0.5^2 = \underline{\hspace{2cm}}$

2. A cube has edges of 3 cm. Its volume is 3^3 cm^3 . Find the volume.*Show your work.***Answer** _____ cm^3 **SUM IT UP***in your own words* 2^4 is not the same as 2×4 because ____.
